

The Fermi-Pasta-Ulam-Tsingou problem

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Consider a horizontal chain of masses connected by springs. Each of them can oscillate in the vertical direction. We will use “horizontal” and “vertical” for clarity, but we are neglecting gravity here.

Hooke’s law tells us that force is related to displacement through $\vec{F} = -k\vec{x}$; for the i -th mass, the horizontal component of this force will always be balanced (since the horizontal distance to its neighboring masses is fixed).

We can parameterize these masses through their vertical coordinates; the i -th mass will be subject to a force

$$F_i = -k(x_i - x_{i-1}) - k(x_i - x_{i+1}) = k(x_{i-1} + x_{i+1} - 2x_i)$$

meaning that its acceleration, through $F = ma$, will be

$$\ddot{x}_i = \frac{k}{m}(x_{i-1} + x_{i+1} - 2x_i)$$

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